The Derivative from First Principles

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The definition of the derivative**

Use the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.

$\frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h} = \frac{2xh + h^2 + 3h}{h} = 2x + h + 3 \to 2x + 3$.

2 **[Foundational]** • **Target: The definition of the derivative**

The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.

a Find the slope of the secant line. $\frac{9-1}{3-1} = 4$.

b Find $f'(2)$ using the derivative, and compare it to the secant slope. $f'(x) = 2x$, so $f'(2) = 4$ — for this quadratic, the secant slope between $x = 1$ and $x = 3$ happens to equal the instantaneous slope at the midpoint $x = 2$.

(a) $\frac{9-1}{3-1} = 4$.; (b) $f'(x) = 2x$, so $f'(2) = 4$ — for this quadratic, the secant slope between $x = 1$ and $x = 3$ happens to equal the instantaneous slope at the midpoint $x = 2$.

3 **[Foundational]** • **Target: Differentiability**

Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.

The left-hand derivative is $\lim_{h \to 0^-} \frac{|h|}{h} = -1$ and the right-hand derivative is $\lim_{h \to 0^+} \frac{|h|}{h} = 1$. Since these disagree, $f'(0)$ does not exist.

4 **[Foundational]** • **Target: Differentiability**

State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?

Differentiability implies continuity (a corner or asymptote would break the limit needed for the derivative). The converse is false — $f(x) = |x|$ is continuous at 0 but not differentiable there.

5 **[Foundational]** • **Target: Interpreting the derivative**

A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.

$h'(2)$ represents the instantaneous vertical velocity of the ball at $t = 2$ seconds. $h'(2) = -10(2) + 20 = 0$ m/s — the ball is momentarily at rest, at its peak height.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Point-derivatives from the definition

Use the alternate form $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3 - 1$.

$f(2) = 7$. $\frac{x^3 - 1 - 7}{x - 2} = \frac{x^3 - 8}{x - 2} = \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = x^2 + 2x + 4 \to 4 + 4 + 4 = 12$: $f'(2) = 12$.

7 [Foundational] • Target: The derivative as the slope of the tangent line

For $f(x) = x^2 - 4x + 1$, use the derivative (found via the limit definition, $f'(x) = 2x - 4$) to find the equation of the tangent line at $x = 3$.

$f(3) = 9 - 12 + 1 = -2$. $f'(3) = 2(3) - 4 = 2$. Tangent line: $y - (-2) = 2(x - 3)$, i.e. $y = 2x - 8$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The derivative as the slope of the tangent line

Explain why the tangent line to $y = x^2$ at $x = 0$ is horizontal, using the derivative $f'(x) = 2x$ found from the limit definition.

$f'(0) = 2(0) = 0$. Since the slope of the tangent line at $x = 0$ is 0, the tangent line is horizontal there — consistent with $x = 0$ being the vertex (minimum point) of the parabola.

9 [Intermediate] • Target: The derivative function versus the derivative at a point

Explain the difference between $f'(x)$ and $f'(3)$, using $f(x) = x^2$ as an example.

$f'(x) = 2x$ is a FUNCTION giving the slope of the tangent line at any point x. $f'(3) = 2(3) = 6$ is a single NUMBER — the specific slope at the point $x = 3$. The function $f'(x)$ can be evaluated at any x to get such a number.

10 [Intermediate] • Target: Estimating a derivative from a table

A table gives $f(1.9) = 8.61$, $f(2.0) = 9.00$, $f(2.1) = 9.41$. Use a symmetric difference quotient to estimate $f'(2)$.

$f'(2) \approx \frac{f(2.1) - f(1.9)}{2.1 - 1.9} = \frac{9.41 - 8.61}{0.2} = \frac{0.80}{0.2} = 4$.

11 [Intermediate] • Target: The derivative of a constant and linear functions

Use the limit definition to show that the derivative of a constant function $f(x) = c$ is 0.

$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = \frac{0}{h} = 0$ for all $h \neq 0$, so $f'(x) = \lim_{h \to 0} 0 = 0$.

12 [Intermediate] • Target: The derivative of a constant and linear functions

Use the limit definition to show that the derivative of a linear function $f(x) = mx + b$ is the constant m.

$\frac{f(x+h) - f(x)}{h} = \frac{m(x+h) + b - (mx + b)}{h} = \frac{mh}{h} = m$ for all $h \neq 0$, so $f'(x) = \lim_{h \to 0} m = m$.

© ClipSAT — clipsat.org | All Rights Reserved

13 [Intermediate] • Target: Interpreting the derivative

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

Step 1 · Implicit Differentiation: Differentiate with respect to x: $2x - \left(1 \cdot y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \Rightarrow 2x - y + (2y - x)\frac{dy}{dx} = 0$. Solve for $\frac{dy}{dx}$: $\frac{dy}{dx} = \frac{y - 2x}{2y - x}$. Step 2 · Condition for Horizontal Tangent: Set the numerator equal to zero while ensuring the denominator is non-zero: $y - 2x = 0 \Rightarrow y = 2x$. Step 3 · Substitute into Original Curve.

14 [Intermediate] • Target: Point-derivatives from the definition

Level 2 · MediumFind the point (x, y) on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

Step 1 · Distance Objective: The distance d from (x, x^2) to $(0, 2)$ is: $d = \sqrt{(x - 0)^2 + (x^2 - 2)^2}$. Minimizing d is equivalent to minimizing the squared distance function $S(x) = d^2$: $S(x) = x^2 + (x^2 - 2)^2 = x^2 + x^4 - 4x^2 + 4 = x^4 - 3x^2 + 4$. Step 2 · Critical Numbers: $S'(x) = 4x^3 - 6x = 2x(2x^2 - 3) = 0$. Critical roots: $x = 0$ and $x = \pm\sqrt{\frac{3}{2}} = \pm\frac{\sqrt{6}}{2}$. Step 3 · Second Derivative Test: $S''(x) = 12x^2 - 6$. • At $x = 0$: $S''(0) = -60$ (certified loc.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The derivative as the slope of the tangent line • Cross-Lesson: 11.01 Sequences and Series

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.

Take natural logarithms of both sides: $\ln y = \ln(x^{\sin x}) = \sin x \cdot \ln x$. Differentiate implicitly with respect to x using the Product Rule on the right: $\frac{1}{y}\frac{dy}{dx} = (\cos x)(\ln x) + (\sin x)\left(\frac{1}{x}\right)$. Multiply both sides by $y = x^{\sin x}$: $\frac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right)$.

16 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 12.01 Convergence Tests

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

Express $y = (x^4 + 3x^2 + 7)^{1/2}$. By the Chain Rule: $y' = \frac{1}{2}(x^4 + 3x^2 + 7)^{-1/2} \cdot \frac{d}{dx}(x^4 + 3x^2 + 7) = \frac{4x^3 + 6x}{2\sqrt{x^4 + 3x^2 + 7}} = \frac{2x^3 + 3x}{\sqrt{x^4 + 3x^2 + 7}}$. Evaluate at $x = 1$: $y'(1) = \frac{2(1)^3 + 3(1)}{\sqrt{1^4 + 3(1)^2 + 7}} = \frac{5}{\sqrt{11}} = \mathbf{\frac{5\sqrt{11}}{11}}$.

17 [Challenging] • Target: Estimating a derivative from a table • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

Compute partial derivatives via Chain Rule: $f_x = \frac{2x}{x^2 + 3y^2}$, $f_y = \frac{6y}{x^2 + 3y^2}$ Substitute $x = 2$ and $y = 1$: $x^2 + 3y^2 = 2^2 + 3(1^2) = 4 + 3 = 7$ $f_x(2, 1) = \frac{2(2)}{7} = \frac{4}{7}$, $f_y(2, 1) = \frac{6(1)}{7} = \frac{6}{7}$

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

First find the components: $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2 - 3$. The first derivative is $\frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3}{2}t - \frac{3}{2t}$. Differentiating with respect to t: $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2} + \frac{3}{2t^2}$. Then $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t}$. At $t = 2$: numerator is $\frac{3}{2} + \frac{3}{8} = \frac{15}{8}$; denominator is $2(2) = 4$. Thus.

19 [Challenging] • Target: The definition of the derivative • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

Here $f(x) = 2x$ and $f'(x) = 2$. The arc length element is $ds = \sqrt{1 + (f'(x))^2}\,dx = \sqrt{1 + 2^2}\,dx = \sqrt{5}\,dx$. The surface area formula gives: $S = 2\pi\int_0^3 f(x)\sqrt{1 + [f'(x)]^2}\,dx = 2\pi\int_0^3 (2x)\sqrt{5}\,dx = 4\sqrt{5}\pi\left[\frac{x^2}{2}\right]_0^3 = 4\sqrt{5}\pi\left(\frac{9}{2}\right) = 18\sqrt{5}\pi$ Verification via cone geometry: The shape is a right circular cone with radius $r = f(3) = 6$ and slant height $l = \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$. Lateral are.

20 [Challenging] • Target: Differentiability • Cross-Lesson: 15.01 Double Integrals

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

Step 1 · Initial Verification: As $x \to 0$, numerator $0 - 0 = 0$ and denominator $0^3 = 0$. Form $\frac{0}{0}$. Differentiating numerator and denominator: $\lim_{x \to 0} \frac{1 - \frac{1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{\frac{(1+x^2)-1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{x^2}{3x^2(1+x^2)}$. Step 2 · Algebraic Simplification: For $x \neq 0$, cancel the common factor x^2: $= \lim_{x \to 0} \frac{1}{3(1+x^2)} = \frac{1}{3(1+0)} = \mathbf{\frac{1}{3}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The definition of the derivative

Tangent line slope to $y = \ln x$ at $x = e$:

a. 1 **b.** e

c. $1/e$ ✓ **d.** $\ln e$

Applying Applications of Derivatives principles, $f' = 1/x$; $f'(e) = 1/e$. Therefore, the verified correct option is $1/e$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Differentiability

The First Derivative Test classifies a critical point c as a local max when f' changes from:

a. positive to positive

b. negative to positive

c. zero to zero

d. **positive to negative ✓**

f increasing then decreasing through c means c is a local maximum.

23 [Foundational] • Target: More derivatives from the definition

A differentiable function $f(x)$ is such that its derivative $f'(x)$ is defined by $f'(x) = 3x^2 - 12x + 9$. The function $f(x)$ has a local maximum at $x = 1$ and a local minimum at $x = 3$. The graph of $y = f(x)$ passes through the point $(0, 5)$. Consider the tangent line to the graph of $y = f(x)$ at the point where $x = 0$. What is the y-intercept of this tangent line?

a. **5 ✓**

b. 12

c. -3

d. 9

The derivative of $f(x)$ is given as $f'(x) = 3x^2 - 12x + 9$. To find the function $f(x)$, we integrate $f'(x)$: $f(x) = \int (3x^2 - 12x + 9)\,dx = x^3 - 6x^2 + 9x + C$. We are given that $f(x)$ passes through $(0, 5)$. Substituting $x = 0$ and $f(0) = 5$ into the expression for $f(x)$: $5 = (0)^3 - 6(0)^2 + 9(0) + C$, which gives $C = 5$. Thus, the function is $f(x) = x^3 - 6x^2 + 9x + 5$. The derivative at $x = 0$ is $f'(0) = 3(0)^2 - 12(0) + 9 = 9$. The point on the graph at $x = 0$ is $(0, f(0)) = (0, 5)$. The equa.

24 [Foundational] • Target: Interpreting the derivative

The slope of $y = 3x - 5$ at any point is:

a. -5

b. 0

c. **3 ✓**

d. undefined

Applying Derivatives principles, Slope of a line $y = mx + b$ is $m = 3$. Therefore, the verified correct option is 3.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Point-derivatives from the definition

Consider the function $f(x) = e^x \sin(x)$. A particle's position on a number line is given by $s(t) = f(t)$ for $t \geq 0$. The particle is momentarily at rest when its velocity is zero. The acceleration of the particle at these moments is given by $a(t) = f''(t)$. Find the smallest positive value of t at which the particle is momentarily at rest, and then determine the acceleration of the particle at that exact moment.

a. **Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\sqrt{2}e^{\frac{3\pi}{4}}$** ✓

b. Rest at $t = \frac{\pi}{2}$, acceleration $a = e^{\frac{\pi}{2}}$

c. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}$

d. Rest at $t = \pi$, acceleration $a = 0$

First, we need to find the velocity, which is the first derivative of the position function $s(t) = f(t) = e^t \sin(t)$. Using the product rule, $v(t) = f'(t) = e^t \sin(t) + e^t \cos(t) = e^t(\sin(t) + \cos(t))$. The particle is momentarily at rest when $v(t) = 0$. Since $e^t > 0$ for all t, we need $\sin(t) + \cos(t) = 0$, which implies $\tan(t) = -1$. The general solutions for this are $t = \frac{3\pi}{4} + n\pi$, where n is an integer. For $t \geq 0$, the smallest positive value occurs when $n = 0$, so.

26 [Foundational] • Target: The derivative as the slope of the tangent line

Consider a function $f(x)$ whose derivative is given by $f'(x) = 6x^2 - 12x + 6$. The graph of the function $g(x) = x^3 - 6x^2 + 9x - 2$ is shown below. If $f(x)$ and $g(x)$ have the same tangent line at $x = 1$, determine the value of $f''(0)$.

a. -6 ✓

b. -12

c. 0

d. 6

First, let's find the derivative of $g(x)$: $g'(x) = 3x^2 - 12x + 9$. At $x = 1$, the slope of the tangent line to $g(x)$ is $g'(1) = 3(1)^2 - 12(1) + 9 = 3 - 12 + 9 = 0$. The problem states that $f(x)$ and $g(x)$ have the same tangent line at $x = 1$. This means $f'(1) = g'(1) = 0$. We are given $f'(x) = 6x^2 - 12x + 6$. Let's check: $f'(1) = 6(1)^2 - 12(1) + 6 = 6 - 12 + 6 = 0$. This confirms consistency. To find the value of $f''(0)$, we first need to find the second derivative of $f(x)$. We are given.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: The derivative function versus the derivative at a point

Find the slope of the tangent line to $r = 2\cos(\theta) + 2$ at $\theta = \frac{\pi}{4}$.

a. $-\sqrt{2}$

b. $1 - \sqrt{2}$ ✓

c. $1 + \sqrt{2}$

d. 1

With $x = r\cos\theta$, $y = r\sin\theta$: $\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$. At $\theta = \frac{\pi}{4}$ this is $1 - \sqrt{2}$.

28 [Foundational] • Target: Estimating a derivative from a table

Parametric tangent at $t = t_0$: slope $m = \frac{y'(t_0)}{x'(t_0)}$. If $x'(t_0) = y'(t_0) = 0$, the point is:

a. always a cusp

b. regular

c. a node

d. **a singular/cusp point — need L'Hôpital** ✓

Applying Parametric and Polar principles, Both zero: singular point. Therefore, the verified correct option is a singular/cusp point — need L'Hôpital.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The derivative of a constant and linear functions

Prove function is constant: if $f'(x) = 0$ on (a, b), then by MVT:

a. $f(x) = f(a)$ **for all** x ✓

b. f is zero

c. f is periodic

d. f is odd

Applying Applications of Derivatives principles, MVT: $f(x) - f(a) = f'(c)(x - a) = 0$. Therefore, the verified correct option is $f(x) = f(a)$ for all x.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: The definition of the derivative

Local min at $x = c$ by First Derivative Test requires f':

a. positive then negative

b. **negative then positive ✓**

c. positive everywhere

d. zero everywhere

Applying Applications of Derivatives principles, Sign change from $-$ to $+$. Therefore, the verified correct option is negative then positive.

31 [Intermediate] • Target: Differentiability

At which point does $f(x) = x^2 - 4x$ have a horizontal tangent?

a. $x = 0$

b. $x = 2$ ✓

c. $x = 4$

d. $x = -2$

Applying Derivatives principles, $f'(x) = 2x - 4 = 0 \Rightarrow x = 2$. Therefore, the verified correct option is $x = 2$.

32 [Intermediate] • Target: More derivatives from the definition

The slope of the tangent to $f(x) = x^2$ at $x = 3$ is:

a. 3

b. **6 ✓**

c. 9

d. 12

Applying Derivatives principles, $f'(x) = 2x$. At $x = 3$: $f'(3) = 6$. Therefore, the verified correct option is 6.

33 [Intermediate] • Target: Interpreting the derivative

The derivative of $f(x) = \frac{x^4 - 1}{x^2 + 1}$ at $x = 1$ using limit definition:

a. 1

b. **2 ✓**

c. 3

d. 4

Simplify: $\frac{(x^2 - 1)(x^2 + 1)}{x^2 + 1} = x^2 - 1$. So $f'(x) = 2x$. At $x = 1$: $f'(1) = 2$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Point-derivatives from the definition

The Second Derivative Test says $f''(c) < 0$ at a critical point c implies:

a. inflection point

b. no conclusion

c. **local maximum** ✓

d. local minimum

Concave down ($f''(c) < 0$) at a critical point means a local maximum.

35 [Intermediate] • Target: The derivative as the slope of the tangent line

Acceleration is the second derivative of position. If $a(t) = g$ (constant), integrating twice gives:

a. $s(t) = \frac{1}{2}gt^2 + v_0t + s_0$ ✓

b. $s(t) = gt$

c. $s(t) = v_0t$

d. $s(t) = g + v_0$

Double integration of constant acceleration gives quadratic.

36 [Intermediate] • Target: The derivative function versus the derivative at a point

The tangent line to $y = \ln x$ at $x = e$ has equation:

a. $y = x/e$

b. $y = x/e - 1 + 1$

c. $y = \frac{1}{e}(x - e) + 1$ ✓

d. $y = e(x - e) + 1$

Slope $= 1/e$, point $(e, 1)$: $y - 1 = \frac{1}{e}(x - e)$.

37 [Intermediate] • Target: Estimating a derivative from a table

The normal line to $y = x^2$ at $x = 1$ has slope:

a. -2

b. $-\frac{1}{2}$ ✓

c. $\frac{1}{2}$

d. 2

Tangent slope $= f'(1) = 2$. Normal slope $= -1/2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The second derivative test: $D = f_{xx}f_{yy} - f_{xy}^2$. If $D > 0$ and $f_{xx} > 0$, the point is a:

a. saddle point

b. **local min ✓**

c. local max

d. inflection point

$D > 0, f_{xx} > 0 \Rightarrow$ local minimum.

39 [Challenging] • Target: The definition of the derivative • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Jerk = third derivative of position. If jerk is constant j, acceleration is:

a. $jt + a_0$ ✓

b. jt^2

c. j/t

d. e^{jt}

One integration of constant jerk.

40 [Challenging] • Target: Differentiability • Cross-Lesson: 15.01 Double Integrals

The graph of a function $g(x)$ is shown above. The function $g(x)$ is defined by $g(x) = \frac{1}{3}x^3 - ax^2 + 2x + 1$, where a is a positive constant. The graph has a tangent line at $x = 1$ which is parallel to the line $y = -2x + 5$. Find the value of the constant a.

a. 1

b. 2

c. **3 ✓**

d. 4

The derivative of $g(x)$ is $g'(x) = x^2 - 2ax + 2$. The slope of the tangent line at $x = 1$ is equal to the derivative of $g(x)$ evaluated at $x = 1$. The given tangent line is $y = -2x + 5$, so its slope is -2. Therefore, $g'(1) = -2$. Substituting $x = 1$ into the derivative: $g'(1) = (1)^2 - 2a(1) + 2 = 1 - 2a + 2 = 3 - 2a$. Setting $g'(1)$ equal to the slope of the tangent line: $3 - 2a = -2$. Solving for a: $-2a = -5$, which gives $a = \frac{5}{2}$. However, upon review, the problem asks for a positive constan.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: More derivatives from the definition • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = 1$ has the equation $y = 2x + k$. Given that $f(1) = 3$, determine the value of k.

a. 1

b. **2** ✓

c. 3

d. 4

The equation of a tangent line to a function $f(x)$ at a point $(a, f(a))$ is given by $y - f(a) = f'(a)(x - a)$. We are given that the tangent line at $x = 1$ has the equation $y = 2x + k$. The slope of this tangent line is 2, which means $f'(1) = 2$. We are also given that $f(1) = 3$. Substituting these values into the point-slope form of the tangent line equation: $y - 3 = 2(x - 1)$ $y - 3 = 2x - 2$ $y = 2x - 2 + 3$ $y = 2x + 1$ Comparing this to the given tangent line equation $y = 2x + k$, we can see that.

42 [Challenging] • Target: Interpreting the derivative • Cross-Lesson: 2.01 Evaluating Limits

The graph of a differentiable function $y = f(x)$ is shown below. The tangent line to the graph at $x = 2$ has the equation $y = -x + k$. Find the value of k.

a. 4

b. **5** ✓

c. 6

d. 7

The graph shows that the function $f(x)$ passes through the point $(2, 3)$. Therefore, $f(2) = 3$. The tangent line to the graph at $x = 2$ also passes through the point $(2, f(2))$, which is $(2, 3)$. The equation of the tangent line is given as $y = -x + k$. Substituting the point $(2, 3)$ into this equation, we get $3 = -2 + k$. Solving for k, we find $k = 3 + 2 = 5$.

43 [Challenging] • Target: Point-derivatives from the definition • Cross-Lesson: 2.02 Continuity

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$ ✓

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

The integrating factor is $\mu(x) = e^{-\int \frac{1}{x}\,dx} = \frac{1}{x}$. Then $\left(\frac{y}{x}\right)' = x \Rightarrow \frac{y}{x} = \frac{x^2}{2} + C \Rightarrow y = \frac{1}{2}x^3 + Cx$.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: The derivative as the slope of the tangent line • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Tangent line to $r = \theta$ at $\theta = \pi$:

a. $y = -x$

b. horizontal

c. **slope π** ✓

d. slope $-\pi$

For $r = \theta$ at $\theta = \pi$, $\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}\big|_{\theta=\pi} = \frac{0-\pi}{-1-0} = \pi$.

45 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Unit tangent vector $\mathbf{T}(t) = \mathbf{r}'(t)/|\mathbf{r}'(t)|$. It has magnitude:

a. 1 ✓

b. 0

c. $|r'|$

d. t

Applying Vectors and Vector-Valued Functions principles, Unit tangent: $|\mathbf{T}| = 1$. Therefore, the verified correct option is 1.

© ClipSAT — clipsat.org | All Rights Reserved